

Ritesh Sur Chowdhury

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About Me

I am currently pursuing an MS by Research in Data Science and Artificial Intelligence at IIT Bombay. Previously, I worked at Qualcomm in 5G communication testing and automation development. I have contributed to multiple research projects, resulting in 11 publications. My coursework spans several domains, including computer vision, natural language processing, reinforcement learning, generative and graphical models. Alongside my coursework, research, and teaching assistant duties, I have actively participated in various competitions.

Research Interests: Deep Learning, Computer Vision, Vision-Language Models, Domain Adaptation, 3D Vision

Education

- **Indian Institute of Technology (IIT), Bombay** July 2024 - Present
MS by Research in Centre for Machine Intelligence and Data Science (CMInDS)
GPA: 9.31/10
- **Jadavpur University** August 2018 - July 2022
B.E. in Electronics and Telecommunication Engineering (ETCE)
GPA: 9.61/10
- **Graduate Aptitude Test in Engineering (GATE)** 2022
Electronics and Communication Engineering
All India Rank: 59, Gate Score: 829/1000

Experience

- **Qualcomm – Hyderabad, India** July 2022 – July 2024
Position: Engineer (December, 2023 - July, 2024), Associate Engineer (July, 2022 - December, 2023)
 - **Team:** FSM System Test (FST), **Project:** Cavalla Sub-6 TDD and FDD (5G)
 - **Job Description:** System Testing (Sub-6 RF) and Automation development (Python)
 - **Awards:** Impact Award

Current research projects:

- Design Resolution-Agnostic MLP-based vision model (Collaboration with University of Massachusetts).
- Pedestrian Trajectory Detection for autonomous vehicles (Collaboration with Honda).
- Single source domain generalization in 3D point cloud.
- Composed Image retrieval.

Publications

* indicates Joint first authorship.

- S. Bose*, **R. S. Chowdhury***, D. Pal, S. Bose, B. Banerjee, and S. Chaudhuri, "Multiscale probability map guided index pooling with attention-based learning for road and building segmentation," *arXiv preprint arXiv:2302.09411*, 2023.
- **R. S. Chowdhury***, S. Bose*, S. Ghosh, and A. Konar, "Attention induced dual convolutional-capsule network (AIDC-CN): A deep learning framework for motor imagery classification," *Computers in Biology and Medicine*, vol. 183, p. 109260, 2024.
- S. Bose*, **R. S. Chowdhury***, R. Das, and U. Maulik, "Dense dilated deep multiscale supervised U-network for biomedical image segmentation," *Computers in Biology and Medicine*, vol. 143, p. 105274, 2022.
- R. Das*, S. Bose*, **R. S. Chowdhury**, and U. Maulik, "Dense dilated multi-scale supervised attention-guided network for histopathology image segmentation," *Computers in Biology and Medicine*, p. 107182, 2023.
- S. Ghosh, S. Bose, **R. S. Chowdhury**, A. Konar, and A. K. Nagar, "Decoding the creative ability of subjects from aesthetic quality assessment using dual convolution induced capsule network," in *Proc. IEEE Symp. Series on Computational Intelligence (SSCI)*, pp. 1319–1326, 2022.

- **R. S. Chowdhury**, J. Jana, S. Tripathi, and J. Bhaumik, “Improved DWT and IDWT architectures for image compression,” *Microprocessors and Microsystems*, p. 104990, 2023.
- **R. S. Chowdhury**, J. K. Sadhu, C. Thakur, and S. Chattopadhyay, “Performance analysis and optimization of a hybrid TSR–PSR protocol for AF, DF and hybrid AF–DF relaying under Weibull fading,” *Telecommunication Systems*, vol. 82, no. 1, pp. 61–90, 2023.
- J. Jana, **R. S. Chowdhury**, S. Tripathi, and J. Bhaumik, “FPGA implementation of compact and low-power multiplierless architectures for DWT and IDWT,” *Journal of Real-Time Image Processing*, vol. 21, no. 1, p. 19, 2024.
- J. Jana, S. Tripathi, A. Bhattacharya, **R. S. Chowdhury**, and J. Bhaumik, “An area efficient VLSI architecture for 1-D and 2-D discrete wavelet transform (DWT) and inverse discrete wavelet transform (IDWT),” in *Proc. Devices for Integrated Circuit (DevIC)*, pp. 378–382, 2021.
- J. Jana, S. Tripathi, A. Bhattacharya, **R. S. Chowdhury**, A. Karmakar, D. Ranjan, and J. Bhaumik, “An Android-application-controlled car for human safety against COVID-19,” in *Internet of Things and Its Applications: Select Proceedings of ICIA 2020*, pp. 3–12, Springer, 2022.
- J. Jana, S. Tripathi, A. Bhattacharya, **R. S. Chowdhury**, D. Ranjan, and J. Bhaumik, “A cost-effective tracking and health monitoring system for suspected COVID-19 patient in quarantine,” in *Microelectronics, Circuits and Systems: Select Proceedings of Micro2021*, pp. 569–579, Springer, 2023.

Course Projects

- Language translation, and change of accent for speech-to-speech task using diffusion model (Course: CS 769)
- Generative AI for Data augmentation in medical images (Course: IE 643)
- Reinventing the Wheel: Fixing Backpropagation by Not Using It at All (Course: CS 725)

Course Work (MS)

- **3rd semester:** CS 772 (Deep Learning for Natural Language Processing), GNR650 (Advanced topics in deep learning for image analysis), SOM727 (Applied Quality Engineering and Management)
- **2nd semester:** CS 726 (Advanced Machine Learning), CS 747 (Foundations of Intelligent and Learning Agents), CS 769 (Optimization in Machine Learning), EE 702 (Computer Vision), GNR638 (Machine Learning for Remote Sensing - II), DS 691 (R & D Project), DS 694 (Seminar)
- **1st semester:** CS 601 (Algorithms and Complexity), CS 725 (Foundations of Machine Learning), EE 601 (Statistical Signal Analysis), EE 635 (Applied Linear Algebra), IE 643 (Deep Learning - Theory and Practice)

Competitions

- **FIFS Gameathon** 2025
1st position in Gameathon1.0 (ICC Champions League)
2nd position in Gameathon2.0 (IPL)
Challenge: Build an ML model to select optimal team of 11 players (Aggregated for a live tournament).
- **Smart India Hackathon (SIH):** Selected for national level (IIT Bombay). 2025
Problem Statement: Optical-Guided Super-Resolution for Thermal IR Imagery
- **Swadeshi Microprocessor Challenge (MeitY):** Secured position in Quarter Final 2020

Teaching Assistanceship

- EPGD, DL and Gen AI Autumn, 2025-2026
- CS791 (Probabilistic Foundations of AI) Autumn, 2025-2026
- DS303 (Introduction to ML) Spring, 2024-2025

Skills

Languages: English, Hindi, Bengali

Programming Languages: C, Python, Matlab

Deep Learning Frameworks: PyTorch, TensorFlow

Hobbies

Chess, Solving puzzles, Playing Badminton